OCTREOTIDE

Variceal bleeding
Glucose changes

CHEMICAL NAME D-Phenylalanyl-L-cysteinyl-L-phenylalanyl-D-tryptophyl-L-lysyl-L-threonyl-N-[(1R,2R)-2-hydroxy-1-(hydroxymethyl)propyl]-L-cysteinamide cyclic (2→7)-disulfide (acetate)
CLASS Somatostatin analogue
TRADE NAMES Sandostatin · Sandostatin LAR (depot) · Mycapssa (oral)

THERAPEUTIC USES / INDICATIONS

- Acute variceal bleeding (adjunct, with endoscopy).
- Acromegaly; carcinoid syndrome and VIPomas (symptom control).
- Malignant bowel obstruction, high-output fistulas (palliative).
- Sulfonylurea-induced refractory hypoglycaemia; prevention of pancreatic surgery complications.

ADMINISTRATION

- **SC** injection (rotate sites; warm to room temperature).
- **IV** bolus/infusion (variceal bleeding).
- **LAR depot:** deep IM gluteal every 4 weeks — by trained staff.
- Store in fridge.

DOSAGES

- **Variceal bleeding:** 50 mcg IV bolus, then 25–50 mcg/h infusion for 2–5 days.
- **Carcinoid/VIPoma:** 50 mcg SC 1–2 times daily, up to 200 mcg three times daily.
- **Acromegaly:** 100–200 mcg SC three times daily; LAR 20–30 mg every 4 weeks.
- **Bowel obstruction:** 300–600 mcg/24 h SC infusion.

PHARMACOKINETICS

- **Onset:** SC 30 min. **Duration:** 6–12 h (LAR 4 weeks).
- Metabolised in liver.
- Excreted in urine and bile. **t½:** 1.5 h.

PHARMACODYNAMICS

- Mimics **somatostatin**: inhibits GH, insulin, glucagon, gastrin, VIP, serotonin.
- ↓ splanchnic blood flow and portal pressure; ↓ GI secretions and motility.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain.
- Nausea, abdominal cramps, diarrhoea, flatulence, steatorrhoea.
- Hyper- or hypoglycaemia.

ADVERSE EFFECTS (SERIOUS)

- Gallstones (long-term).
- Bradycardia, conduction abnormalities.
- Hypothyroidism.
- Pancreatitis; vitamin B12 deficiency.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** diabetes (adjust insulin), gallbladder disease, cardiac conduction disorders.

KEY INTERACTIONS

- **Insulin, sulfonylureas:** dose adjustment needed.
- **Ciclosporin:** ↓ absorption.
- **β-blockers, calcium-channel blockers:** bradycardia.

NURSING MANAGEMENT

- Variceal bleed: monitor BP, HR, Hb, signs of rebleeding.
- Monitor blood glucose regularly.
- Rotate SC sites; let vial reach room temperature.
- Long-term: gallbladder ultrasound, TFTs, ECG.

PATIENT EDUCATION

- Injection-site discomfort is reduced if the drug is at room temperature.
- Monitor glucose if diabetic.
- Report severe abdominal pain or yellow skin.

KEY POINTS

- Somatostatin analogue.
- Carcinoid, acromegaly.
- Variceal bleed: 50 mcg bolus + 25–50 mcg/h.
- Glucose changes; gallstones.